

COSC 3P32 – Introduction to Database Systems

Winter 2026

Group Project, part 1

Due Date: March 10th, 2026, 11:59pm

No late assignments are accepted.

This work accounts for 4% of your final grade and is worth a total of 40 marks.

There are 2 parts to this project. In part 1 of the project, you are to revise the design of the video streaming service database already seen in Assignment 1. In part 2, you are to use database tools to build the database, and demonstrate different ways of interacting with the database.

The information to be stored in the database is the same as given in assignment 1, with the following exceptions and additions:

- In the interval during which a viewer watches a video, the start of the interval must be no later than the end of the interval. Also, viewers are not able to watch more than one video at the same time (i.e. the intervals for watching the video cannot overlap).
- The year of a video must be between 1900 and 2026.
- The rating of a movie must be an integer between 0 and 100.
- Every show must have at least one season. For each show with s seasons, the episodes for that show must have a season number between 1 and s .
- The attributes “account type”, “monthly cost”, and “credit card” must be added to the Account table. The same credit card cannot be used by more than one account. There are two possible account types (Standard and Premium), and all accounts of the same type have the same monthly cost. Standard accounts can only be used by 1 viewer, while Premium accounts can be shared by any number of viewers.

Note that some of the above information will lead to functional dependencies within a table, and must be dealt with in part 1 of the project. Others will require the use of check constraints, triggers or other means, and must be implemented in part 2 of the project. There may also be constraint(s) that could not be handled in assignment 1 because you did not have the required knowledge at the time – these also must be handled in part 2 of the project.

Note: Sample solutions (including an ER diagram and a set of `CREATE TABLE` statements) for assignment 1 are available on Brightspace. You can either start this project using that solution, or modify your own solution, as you wish.

For part 1 of your project, you must complete all of the following steps:

1. [4 marks] Provide an ER model for the above database. All attributes of each entity set and relationship set must be shown. All keys must be shown for each entity set. All key and participation constraints must be shown for each relationship set. In addition, write a paragraph specifying any necessary overlap and covering constraints.
2. [4 marks] Based on the ER model, specify a relational schema for the database.
3. [24 marks] For each relation, identify all functional dependencies that hold on the fields of that relation and specify if it is in BCNF, 3NF or neither. If a relation is not in BCNF, then attempt to find a BCNF decomposition that is both lossless-join and dependency-preserving. If this is

not possible, then a lossless-join, dependency-preserving 3NF decomposition is acceptable. Clearly specify the resulting relational schema.

4. [8 marks] Write the SQL statements necessary to create the tables for the above database, capturing as many constraints as possible. Note: the implementation of check constraints, triggers, etc. can be delayed until part 2 of the project and no marks are allocated for these in part 1.

Submission Requirements:

1. Ensure that you have fully complied with all notation and formatting requirements.
2. Each group must submit a report containing all answers to all questions.
3. The report should also have a cover page that includes your lab number, group number, and the names and email addresses of all members of the group.
4. The report must be submitted as a single PDF file per group, using Brightspace.